CSF306: Mobile App Development – Coursework 2

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Due: 11:00am, 25th March 2026

Task

This coursework requires you to develop the Flutter app you previously designed in coursework 1. You are expected to spend time each week incrementally developing your app from the content you have learned that week. Throughout the development process, you are expected to develop professionally, keeping track of the work you have done and the code you have written.

The app itself should adhere closely to your initial design, with no major changes unless discussed during feedback. You will be graded on the fullness and completeness of your app, as well as how professional and purposeful it is. You will score highly for a well-designed, meaningful app, compared to something which feels rushed and incomplete. Due to the timescale of this coursework, it is highly recommended that you do small increments of development each week, to allow you to fully think and develop each feature of your app.

Alongside having features you designed in Coursework 1, there is a list of things you are recommended to include from the content taught in this module. Below is a complete list of the things taught throughout the 10-week course, and **it is expected you include at least 8 out of the 15** topics discussed:

- Widgets
- Various Media Assets
- Navigable Screens
- Dialog Boxes
- Data Persistence Across Screens
- Menu's
- API Calls
- Firestore
- Forms
- Device Features
- Launching URLS
- App Gestures
- Animations
- Testing
- Accessibility

Naturally, not all of them need to be completed, and it's important to ensure that the ones you do include are meaningful and not just included for the sake of it. Please discuss with the module leader if you feel there are features you can't include, and are struggling to find a place to place them meaningfully.

Generative A.I

When developing your app, you are permitted to use all standard tools available to you, including Generative A.I. (i.e., ChatGPT, Copilot, etc.). When doing so, I have strict guidelines on how to declare any code generated:

- Place opening and closing Dart comments around any code generated by Generative A.I tools
- Place a comment discussing how much (if any) modification you made to the code

Marking

The application will be marked according to the rubric on the following page. Each one of the requirements will be scored individually, as well as points given for a cohesive and easy-to-use app, consistency with the original design, and good justification during the **10-minute video**.

In total, you will be required to:

1. Submit your **10 minute video** which **include answers to questions** as follows (where applicable):
   - Can you explain the core purpose of your app and how it aligns with your initial design from Coursework 1?
   - What challenges did you face in implementing your app, and how did you overcome them?
   - Which 8 out of the 15 required features did you choose to include, and why?
   - Can you walk me through the structure of your app, including its main widgets and navigable screens?
   - How does your app handle data persistence across screens? Can you show an example?
   - How did you integrate media assets into your app?
   - If your app includes API calls, how did you ensure they work efficiently and securely?
   - Did you implement Firestore? If so, how are you managing data within it?
   - What device features (e.g., camera, GPS, notifications) does your app use, and how are they implemented?
   - How does your app handle user inputs via forms, and what validation mechanisms are in place?
   - Have you incorporated animations or gestures to enhance the user experience? Can you demonstrate?
2. Provide a **single zip file** containing the app itself. A Zip file of the app must be provided as proof of completion, although it will not be used for the majority of the marking, only for corroboration and additional marks for professional code.

By submitting coursework, electronically and/or hardcopy, you state that you fully understand and are complying with the University's policy on Academic Integrity and Academic Misconduct. The policy can be found at `https://myuni.swansea.ac.uk/academic-life/academic-misconduct`*. The consequences of committing academic misconduct can be extremely serious and may have a profound effect on your results and/or further progression. The penalties range from a written reprimand to cancellation of all of your marks and withdrawal from the University.*

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Criteria	Ratings						Pts
App Design How well designed is the app, does it align with design guidelines? How simular is it to the proposed app?	**20 to >16 Pts** **Full marks** The app is well designed, with justification relating to design principles.	**16 to >12 Pts** **First (70%-79%)** The App is reasonably well-designed but lacks justification	**12 to >8 Pts** **2:1 (60-69%)** The apps design is done well, but lacks any reference to design principles	**8 to >4 Pts** **2:2 (50-59%)** The app's design is fine, but a bit bland or basic	**4 to >0 Pts** **Pass (40-49%)** The app's design is functional, but not much else.	**0 Pts** **No marks**	20 pts
App Functionality and Purpose How well does the work and run, is it intuitive, does it work?	**35 to >28 Pts** **Full marks** The app functions well, isn't missing any features, and works intuitively.	**28 to >21 Pts** **First (70%-79%)** The app functions well, is only missing small features, and works intuitively.	**21 to >14 Pts** **2:1 (60-69%)** The app functions okay, is missing small features, and works fine, although not obvious.	**14 to >7 Pts** **2:2 (50-59%)** The app functions poorly, and is missing features from the original design. Additionally, the app doesn't work as expected.	**7 to >0 Pts** **Pass (40-49%)** The app functions very poorly, missing most features from the original design, and doesn't work at all as what would be expected.	**0 Pts** **No marks**	35 pts
Required App Features Does the app contain all necessary features?	**30 to >24 Pts** **Full marks** The learner has integrated over 10 features, at least 8 of which are done well.	**24 to >18 Pts** **First (70%-79%)** The learner has integrated 8, well thought out features.	**18 to >12 Pts** **2:1 (60-69%)** The learner has integrated 8 features, but only 6 have been through out well.	**12 to >6 Pts** **2:2 (50-59%)** The learner has integrated 8 features, less than half have been through out well.	**6 to >0 Pts** **Pass (40-49%)** The learner has integrated less than 8 features.	**0 Pts** **No marks**	30 pts
Quality of Code How professional is the code, is it industry-standard?	**5 to >3 Pts** **Full marks** The code is written to a professional standard, with comments.			**3 to >0 Pts** **2:1 (60-69%)** The code is written well, but lacks comments.		**0 Pts** **No marks**	5 pts
Quality of video How well did the learner complete their video? Did they answer questions well?	**10 to >7 Pts** **Full marks** A well-presented and answered viva	**7 to >4 Pts** **2:1 (60-69%)** A well presented viva, with some questions unanswered		**4 to >0 Pts** **Pass (40-49%)** A reasonable viva with mostly fine answers		**0 Pts** **No marks**	10 pts
							Total points: 100